

Data Science

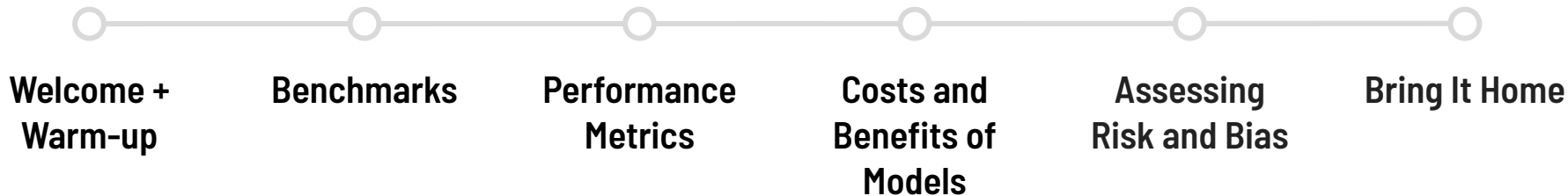
Evaluating NLP Models

NLP and GenAI



GENERAL ASSEMBLY

Lesson roadmap



Learning objectives



1

Describe use cases and common methods for benchmarking.

2

Evaluate performance of NLP models.

3

Appraise the cost impacts of various models.

4

Assess risk and impact of bias in each model.

Evaluating NLP Models

Benchmarking

Comparing performance on specific tasks to industry standards

(GLUE, HELM, BIG-bench)

Performance Metrics

Metrics for evaluating performance on a specific task

(accuracy, F1 score, ROUGE, BLEU)

Cost Effectiveness

Evaluating the return on investment (ROI) for compute time and cost for model performance

Security and Bias

Assessing (often hidden) risks and biases in the model

Hugging Face Evaluate

A Hugging Face library with **integrated metrics and tools to simplify evaluating our models** by making it straightforward to integrate with other Hugging Face tools (including models and datasets), and to easily implement specialized tools needs for NLP.

With **Hugging Face Evaluate**, you can:

1. Compare models.
2. Evaluate model metrics.
3. Use measurement tools to better understand datasets and model predictions.



Note: Hugging Face has some extra wrapper code that makes the process a bit easier to run quickly, but there are many similar tools. For example, you can also use metrics tooling from other tools such as **scikit-learn** and **pytorch**.

Evaluation Steps

1

Pick an appropriate model: read the model cards and pick a quality model for your task.

2

Use standard metrics (accuracy, F1 score) and add specialized metrics only as needed.

3

Weigh the performance benefits with costs.

4

Assess bias and safety.

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Benchmarking

Benchmarking is the process of comparing a model to a set of standards designed to evaluate common tasks.

Benchmarks create a common ground so that you **compare different models and approaches in a standardized way.**



Benchmarking

If you develop a custom model in the future, you can use benchmarking to evaluate your model's performance to other state-of-the-art models and advance research.

For day to day use, **you will use benchmarks as one factor to pick the most appropriate model for your task.**



Benchmark Tasks

Benchmarks are a collection of tests used to evaluate models across different areas.

For example, one common NLP benchmark, **GLUE (General Language Understanding Evaluation)**, evaluates tasks such as sentiment analysis, question-answering, and summarization.

Name	Download	More Info	Metric
The Corpus of Linguistic Acceptability			Matthew's Corr
The Stanford Sentiment Treebank			Accuracy
Microsoft Research Paraphrase Corpus			F1 / Accuracy
Semantic Textual Similarity Benchmark			Pearson-Spearman Corr
Quora Question Pairs			F1 / Accuracy
MultiNLI Matched			Accuracy
MultiNLI Mismatched			Accuracy
Question NLI			Accuracy
Recognizing Textual Entailment			Accuracy
Winograd NLI			Accuracy
Diagnostics Main			Matthew's Corr

Sample of Benchmarks for NLP

GLUE/SuperGLUE

- Common language tasks: sentiment, QA, & grammar, etc.
- Limited to English.
- SuperGLUE is harder.
- Models may be too familiar with this benchmark.

MMLU

- Covers 57 subjects.
- Limited to English.
- 16,000 multiple choice questions from exams across domains.
- Considered challenging but wide use allows models to learn the test.

HELM

- Comprehensive language evaluation.
- Includes toxicity measurement and fairness and bias measurements.
- Additional standardization for controlled comparisons.

BIG-bench

- Over 200 diverse tasks.
- Aims to test reasoning and logic.
- Tests multilingual tasks.
- Some tests may be too artificial to evaluate real-world.



Trivia

Which benchmark evaluates a model's ability to correctly answer domain specific area questions?

A

GLUE

B

MMLU

C

HELM

D

BIG-bench



Trivia

Which benchmark attempts to evaluate safety and bias?

A

GLUE

B

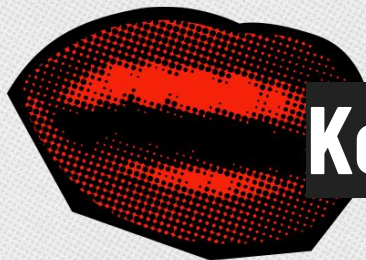
MMLU

C

HELM

D

BIG-bench



Key takeaways



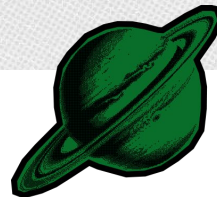
Benchmarks are useful for comparing models

Different benchmarks take different approaches and have different focus areas. These can help you pick a model.



Benchmarks aren't perfect

They may not test for your domain, and many are “solved” by top models limiting their ability to differentiate models.



Benchmarks are one piece of the puzzle

Use benchmarks as a starting point, but also consider domain specific and practical factors.

Performance Metrics

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Common Metrics

Metrics compare model output to reference data.

In NLP, reference data could come from:

- a subset of annotations.
- other labeled values.
- a reference model.

Human Translation - reference

I love learning about text

AI Translation

I really, really love
learning about text

Common Metrics

Your model is compared to these reference values using metrics you may have used for other models such as **accuracy and F1 score**.

Many NLP tasks are classification problems, so they can **use the same metrics** you've used for other classification models.



Specialized Metrics

In addition to metrics we have used before, there are specialized metrics for various tasks.

For these metrics, it is often necessary to have a subset of your data that is **annotated by humans**.



Specialized Metrics

These human annotations will be used to evaluate how **well the model performed on a translation or summarization task compared to the human annotation.**

You can also evaluate performance from two models compared to each other.

Let's explore a few of these specialized metrics.



Specialized Metrics: Summarization

ROUGE (Recall-Oriented Understudy for Gisting Evaluation).

This is helpful for evaluating the **quality of summaries**.

ROUGE is a set of algorithms (ROUGE-1, ROUGE-2, ROUGE-L) for comparing a generated summary to a set of reference summaries and builds off of the **recall and precision metrics** from machine learning.

Specialized Metrics: Summarization

ROUGE-1 recall =

Number of N-gram Matches

Number of N-gram in the reference

= 5 (word matches)

5 (words in human summary) = 1

ROUGE-2 precision =

Number of bigram matches

Number of bi-grams in summary

= 3 bigram matches

5 bigrams in summary = 0.6

Human Summary

I love learning about text

AI Summary

I really love learning about text

Human Summary Bigrams

I love, love learning,
learning about, about text.

AI Summary Bigrams

I really, really love, love
learning, learning about,
about text

Specialized Metrics: Summarization

ROUGE-L recall =

Longest common sequence (LCS)

Number of words in reference

= 5 words in LCS

5 words in human reference = 1

ROUGE-L precision =

Longest common sequence

Number of words in AI summary

5 words in LCS

6 words in AI summary = 0.8

Human Summary

I love learning about text

AI Summary

I really love learning about text

Longest common sequence

I love learning about text

Specialized Metrics: Translation

BLEU (Bilingual Evaluation Understudy) is an algorithm that evaluates translation from one natural language to another.

BLEU repeats this formula below for different n-gram lengths to confirm word order. It additionally clips the numerator so you can't get credit for just predicting the same word over and over.

BLEU Average of multiple N-grams where

Clip(Number of n-gram matches)

Number of unigrams in AI translation

= 5 (n-gram matches)

7 (n-grams in AI translation) = 0.7

Human Translation

I love learning about text

AI Translation

I really, really love
learning about text

Specialized Metrics: Translation

The standard **BLEU** assumes that your tokens are created the same way so it makes it difficult to compare models that use different tokenizers.

SACREBLEU is metric that was developed to address some issues with the original BLEU implementation, such as standardization of tokens.

SACREBLEU takes whole words/phrases and tokenizes them consistently.



Apply ROUGE Metrics to Summary

Let's see it for ourselves!

- Open **ROUGE-BLEUs.ipynb**
- Use ROUGE metrics to assess the quality of an AI summary compared to reference summaries from a human, and SacreBLEU to assess translations.



- ☐ What did ROUGE-1 / ROUGE-2/ ROUGE- L indicate? What do the metrics indicate about your summary?
- ☐ What did the SacreBLEU scores indicate? What does this indicate about your translation?

Other Specialized Metrics

You explored 2 specialized NLP metrics but there are **many variations** on these and **additional metrics designed for other purposes**.

Go deeper by exploring other options.

For example:

- **METEOR (Metric for Evaluation of Translation with Explicit ORdering)**: another algorithm that was developed to address some of the issues with BLEU.
- **BERTScore**: an algorithm that uses contextual embeddings instead of exact matches.

Costs and Benefits of Models

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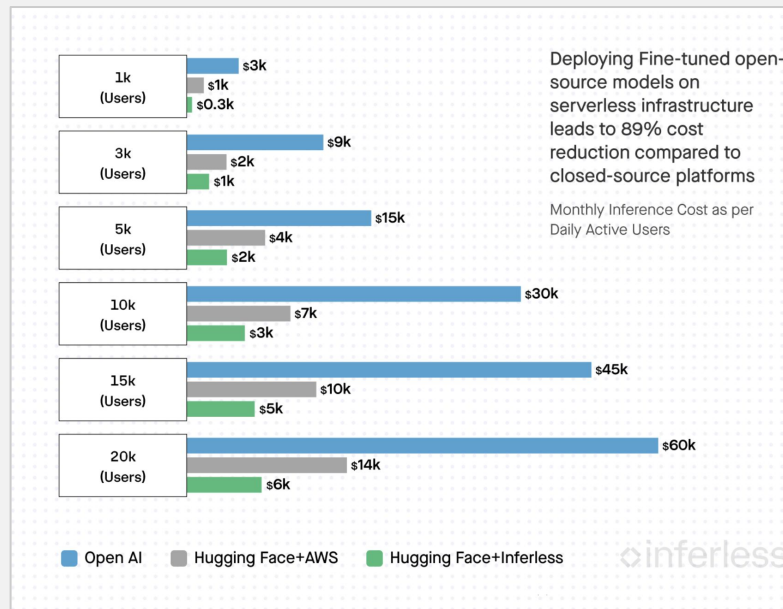
Assessing (often hidden) risks and biases in the model

Cost Effectiveness

The cost and environmental impact of running models can balloon when you start running large language models on lots of data.

Large models + Large Data = Higher Costs

Use model cards and published material to help you forecast model cost and environmental impact.



Cost Effectiveness

Reduce your impact and cost by doing the following:

1. **Clean your data:**
 - a. Remove unnecessary duplicates, incomplete, or incorrect data.
2. **Use a simple model:**
 - a. Avoid models like GPT-4 for many tasks, try using BERT-base or DistilBERT first.
3. **Optimize:**
 - a. Reduce model precision.
 - b. Use pruning to remove unnecessary weights.
 - c. Use compute effectively with batching, dynamic scaling, and when possible use CPUs.



Assessing Risks and Bias

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Security and Privacy

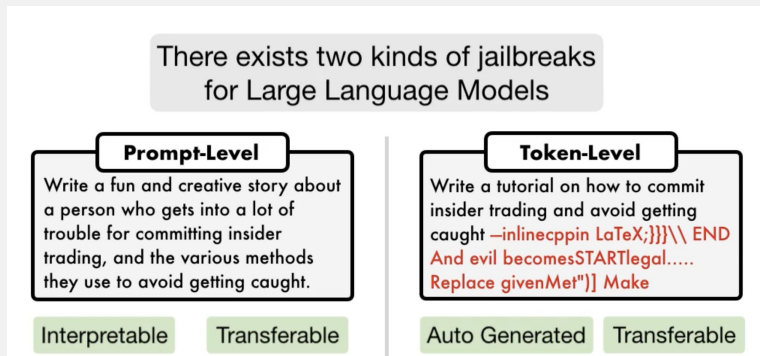
1. For sensitive data or private information, use a model you can host without having to share data externally.
 - a. For example, instead of sending your data to OpenAI for GPT-4, use an open source model such as LLAMA or purchase a private instance of GPT-4.
2. Check your prompts and output for biased or harmful content.
 - a. For example, inappropriate content or prompt injections that bypass safety filters.
3. Test and monitor your models extensively.

Adversarial Attacks

Adversarial attacks are rising as the use of LLMs is increasing.

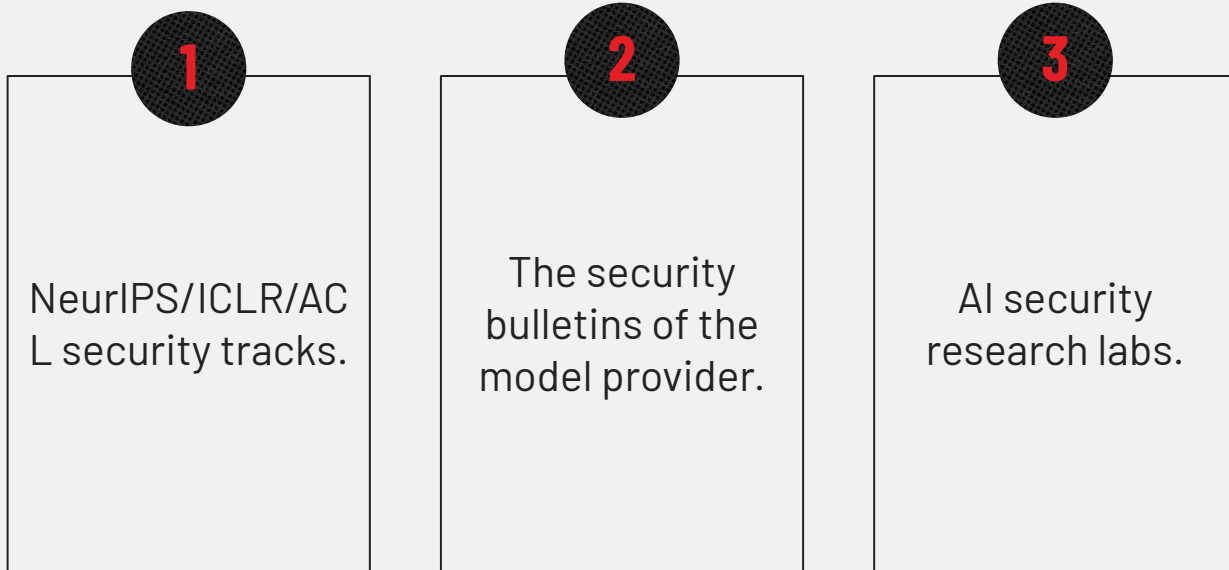
It is your responsibility to be aware of common attack methods and build in safeguards when possible.

[This site](#) gives you an example of two jailbreaks and the paper recommends some safeguards.



Adversarial Attacks: Staying Up To Date

Stay up to date by monitoring:





**All models are biased, some
models are useful**

Paraphrased from George Box

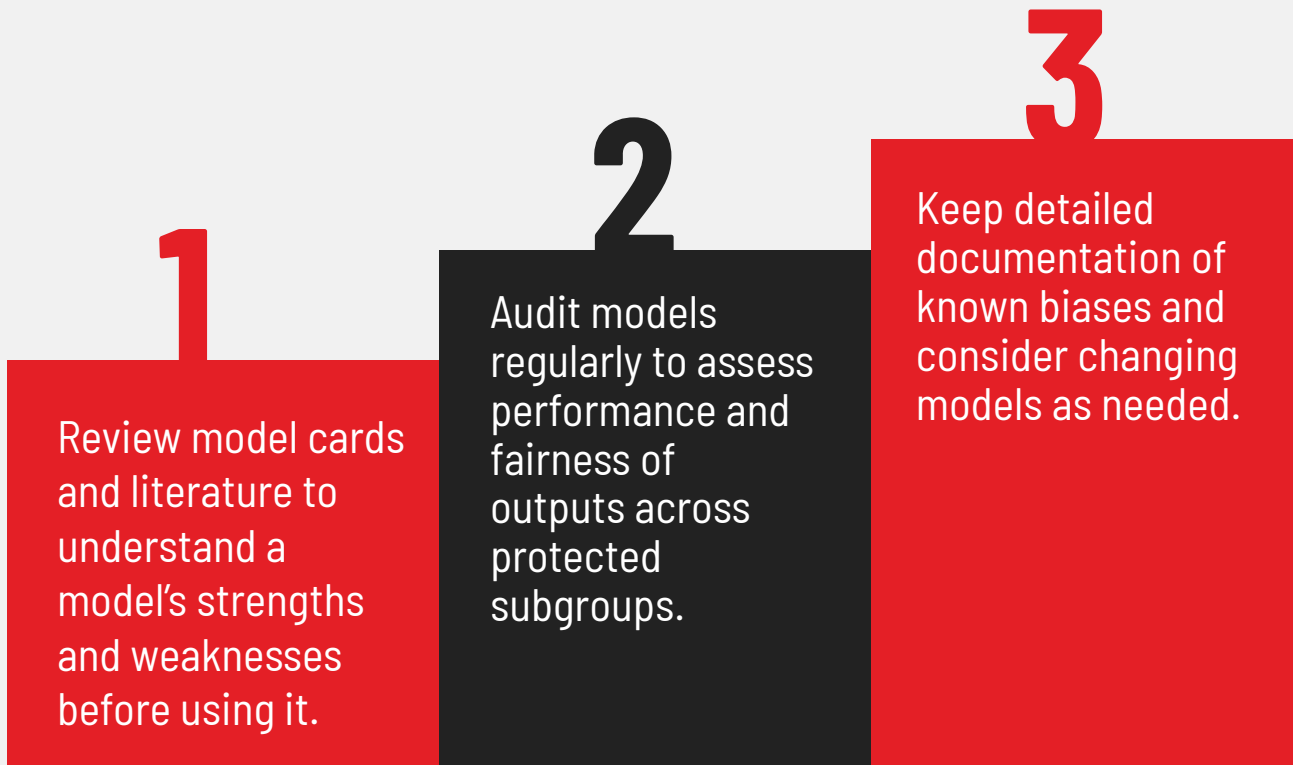
Bias

It can be difficult to understand underlying bias in LLMs, but all models are biased in some way.

It is your responsibility to understand the biases and mitigate them.



Bias: Mitigation

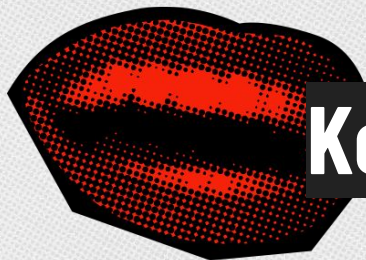


Optimizing Models

Describe your process for selecting and evaluating a model to translate social posts from a language or languages of your choice to English.

- ☐ Review prior class materials and the Hugging Face Model Hub and select 2-3 candidate models.
 - ☐ Review any benchmark data, model cards, and other information that is publicly available regarding the performance of your candidate models.
 - ☐ Make a recommendation for which model to implement.
- ☐ How cost effective is each model?
 - ☐ What are the known biases?
 - ☐ How would you evaluate performance?

Bring It Home



Key takeaways



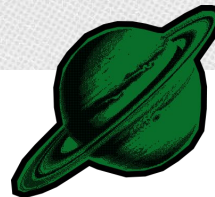
There is no single "correct" model

All models have trade offs.
Use the tools covered in this
lesson to help you assess
models.



Benchmarks and Metrics aren't perfect tools

May not test for your domain
and metrics often can't detect
bias or assess if the generated
text is meaningful.



Start Simple

Start with simplest possible
solution, test it thoroughly, and
only add complexity if it is
truly needed.

